Digital Image representation:

Intensity Resolution: (L)

Law: $L = 2^k$ Where: L: no. grey levels
K: bit depth (bits/pixel)
Note: L interval or (range):
[0, L-1]
يعني مدى L من 0 لـ 255

Spatial Resolution:

$\Rightarrow = M \times N$ Where: M: no. rows
N: no. columns

Dynamic Range:

= Max - Min OR min : Max

Digital Image Storage:

Image Size(b)

$= M \times N \times K$ in bits Where: L: gray level.
K: bits/pixel
M: no. rows
N: no. columns

OR

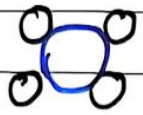
$= M \times N \times \log_2(L)$ in bits

Relation Between Pixels:

☆Concept:

→ Neighbors (N_4, N_8, N_D): أنا هنا جوه بعرف الـPixel اللي حواليا

N_4:  Pixels share edge

N_D:  Pixels share corner

N_8:  Pixels share edge + corner

☆Apply concept:

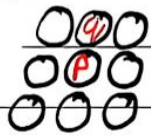
→ Adjacency:

Neighbors was only for Position, BUT adjacency needs a condition:

Two Pixels are adjacent if:

1. They are neighbors (N_4, N_8, or N_D)
2. AND their values are in the connectivity set(V)

Types of Adjacency:

4-adjacent:  q is 4-Adjacent to P, if $q \in N_4(P)$ AND $\{q, P\} \in V$

8-adjacent:  q is 8-Adjacent to P, if $q \in N_8(P)$ AND $\{q, P\} \in V$

m-adjacent: It removes ambiguity from 8-adjacent

"Unique path"

RULE:

q is m-adjacent to P if:

(1) $q \in N_4(P)$ AND $\{P, q\} \in V$

OR

(2) $q \in N_D(P)$ AND $\{q, P\} \in V$ AND
$N_4(P) \cap N_4(q) \cap V = \phi$

Case (1): لو الـ q في N_4 وجوا الـ V فخلاص هي m-adjacent على طول

Case(2): هنا احنا قولنا الـ m-adjacent بـ remove الـ ambiguity يعني هو بلوقتي عايز يشوف هل الـ q اللي في الـ diagonal مع الـ P هتبقى m-adjacent ولا لأ؟

1- لو الـ N_4 بتاعت الـ P و q مفيش حاجة مشتركة يبقى m-adjacent

2- لو طلع في N_4 مشتركة يبقى مش m-adjacent

يعني من الآخر هو بيبدي أولوية للـ N_4 قبل الـ N_D

Image Compression:

☆ Image Compression (image representation) { less transfer time / less space storage

To measure compression ratio CR =:

$$CR = \frac{S_1}{S_2} = \frac{\text{Size before compression}}{\text{Size after compression}} \geq 1$$

if $CR = 1$, : means no compression achieved
if $CR < 1$, : means Stretching happened

Types of Image Compression:

☆ Lossy → original ≅ Decompressed

☆ Exact → original = Decompressed => Huffman Encoding

Huffman Coding Compression algorithm Steps:

1. Histogram
2. Start Sorting with lowest Probability from biggest to lowest
3. Combine 2 lowest Probabilites
4. repeat 2 until all probabilites are combined

Note: بعد كل تجميعة خد بالك ترتب الـ Prob تاني من الكبير الصغير

∴ Calculate Lavg (average length of code):

=

= Probability x (no. bits of that Probability) in bits

$$\therefore CR = \frac{\text{no. bits of image before compression}}{\text{average length of code}} = \frac{K}{L_{avg}}$$

Image Compression Criteria:

CR: how much image size is reduced after compression

RMSE: Measures the difference between the Source image and compressed image Lower is better perfor

$$= \sqrt{\frac{1}{MN} \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} (f(i,j) - f'(i,j))^2}$$

PSNR: Measurs the image quality relative to error introduced: Higher is ~~bad~~ less noise

$$= 10 \times \log_{10} \left(\frac{(L-1)^2}{MSE} \right)$$

Question 3:

a) grey level low/high

Grey level(x)	Count	P(x)	CDF	Equalized: (L-1)*CDF
109	400	10/21	10/21	121
121	250	25/84	65/84	197
128	10	1/84	11/14	200
141	45	3/56	47/56	214
146	70	1/12	155/168	235
159	65	13/168	1	255
Total:	840=MxN	1		

b) Huffman

Original Source			Source reduction			
Grey level(x)	Count	P(x)	1	2	3	4
109	400	0.41	0.41	0.41	0.41	0.41
121	250	0.201	0.201	0.201	0.201	0.41
146	70	0.088	0.088	0.159	0.201	
159	65	0.077	0.077	0.088		
141	45	0.085	0.064			
128	10	0.01				

	Code
0.4	1
0.2	01
0.088	001
0.07	0000
0.05	00010
0.01	00011

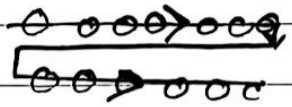
$L_{(avg)} = (0.4)(1)+(0.2)(2)+(0.08)(3)+(0.07)(4)+(0.05)(5)+(0.01)(5)$

$= 1.62$ bits/symbol

$\therefore CR = 4.93 = \frac{K}{L_{avg}}$

Connected component Labeling

Algorithm: (4-adjacency)
Scan Left → Right, Top → Bottom



Anything outside V=0
else
  Check Left and above neighbors
  if no neighbor has label >0, make new label
  if has label >0 ∴ take that label
  if both have different labels
    take either
    Record label A = label B (equivalent)

Distance Measures:

Euclidean distance (D_e)

$$\text{Law: } \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

City Block distance (Manhattan distance) (D_4):

$$D_4 = |x_2-x_1| + |y_2-y_1|$$

Chessboard distance (D_8):

$$D_8 = \max(|x_2-x_1|, |y_2-y_1|)$$

Path distance (D_m):

D_m = length of shortest m-adjacency path

Contrast Stretching (Maps narrow range to wider range)

Basic → like normalization

$$S = \left[\frac{(r - r_min)}{(r_max - r_min)}\right] \times (L-1)$$

Variables:
r_min: lowest intensity
r_max: highest intensity

Piecewise

Variables:
$(r_1, S_1), (r_2, S_2)$: control points
r = input pixel intensity
S = output pixel intensity

$$\alpha = \frac{S_1}{r_1}, \quad \beta = \frac{S_2 - S_1}{r_2 - r_1}, \quad \gamma = \frac{((L-1) - S_2)}{((L-1) - r_2)}$$

$$S = \begin{cases} \alpha \times r & , 0 \le r \le r_1 \\ \beta \times (r - r_1) + S_1 & , r_1 \le r \le r_2 \\ \gamma \times (r - r_2) + S_2 & , r_2 \le r \le L-1 \end{cases}$$

Image Negative: (Enhances white/gray details in dark regions)

$$S = (L-1) - r$$

Lograthimic transform: Expands dark values, compresses bright values
Enhances details in dark regions

$$S = c * \log(1 + r) \quad ; c = \frac{(L-1)}{\log(L)}$$

Power Law (Gamma Correction)

$\gamma > 1$: Darkness image
$\gamma < 1$: Brighter image (expands bright, compress dark)
$\gamma = 1$: No change

$$S = c \times r^{\gamma} \quad ; \quad \frac{(L-1)}{(L-1)^{\gamma}}$$

Grey level Slicing: Highlights specific intensity range

$$S = \begin{cases} L-1 & A \le r \le B \\ 0 & \text{otherwise} \end{cases} \quad OR \quad S = \begin{cases} L-1 & A \le r \le B \\ r & \text{otherwise} \end{cases}$$

Preserve Background

Bit-Plane Slicing: (used for compression)

Morphological Processing

↳ It works only on a Binary Image

Concept:

let Img B => SE ; Img B is the mask.

* Structuring Element (SE) :=>

1	1	1
1	1	1
1	1	1

★ باخد الـ SE مش شرط تكون 3x3 و واحدينها على الصورة عندي زي الـ kernel

3 cases:
- * fit => Match all
- * hit => One match
- * Miss => No match

وبلاقي ٣ حالات:

١- الـ mask كله مطابق fit

1	1	1
1	1	1
1	1	1

Filters (Usage)

1- * Erosion $A \ominus B$ ↗ like min

أدور على اني كل أو أكتر الصورة fit فتتفق

- 1 B fits A
- 0 otherwise

Application: it shrinks objects

٢- الـ mask على الأقل واحد مطابق hit

0	0	0
0	1	0
0	0	0

٣- الـ mask ولا واحدة مطابقة Miss

0	0	0
0	0	0
0	0	0

2- * Dilation $A \oplus B$ ↗ like max

عشان أعرف أفرق ما بينهم

أدور على الـ hit أكتر الصورة فتتفتح

- 1 B hit A
- 0 otherwise

Application: it enlarges objects

ملحوظة: على عكس بقية الماسكات اللي

كانت دايما nxn حيث ان n دايما رقم odd هنا في الـ Morphological Processing الـ mask ممكن يكون أي

حاجة mxn أو حتى كان m4 وهطنش الـ Corners

	1	
1	1	1
	1	

I can use these 2 filters in mix:

★ Opening: remove noise / Preserve Shape / remove bright noise

$A \circ B = (A \ominus B) \oplus B$ دايرة open بفتح بعدها بقفل

★ Closing: fill the holes / Smooth the edges / preserve Shape / remove dark noise

$A \bullet B = (A \oplus B) \ominus B$ دايرة Closed بقفل بعدها بفتح

★ Gradient: Sharp change in intensity (highlights)

$G = (A \oplus B) \ominus (A \ominus B)$